UNIT 1

Teacher Resources

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Name _____ Date _____

Activity Page 1.1 Use with Lesson 1.

Energy Scavenger Hunt

Change is all around you! **Energy** causes changes. Explore the space you are in. Look for things that change in some way. (Hint: Almost everything can!)

Find three examples of things that change. Complete the table to describe your choices.

Three samples have been done for you.

Something That Changes	How Does It Change?
a light bulb	It changes from not bright to bright.
an electric fan	It changes from not moving to moving.
the grass outside	It changes from short to tall.

Write two questions that you have about your examples.

1. _____

_____

2. _____

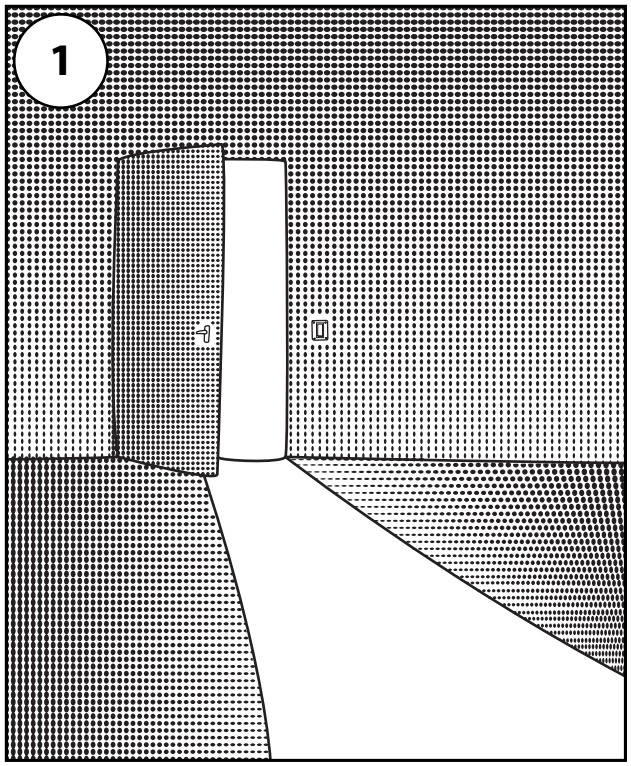
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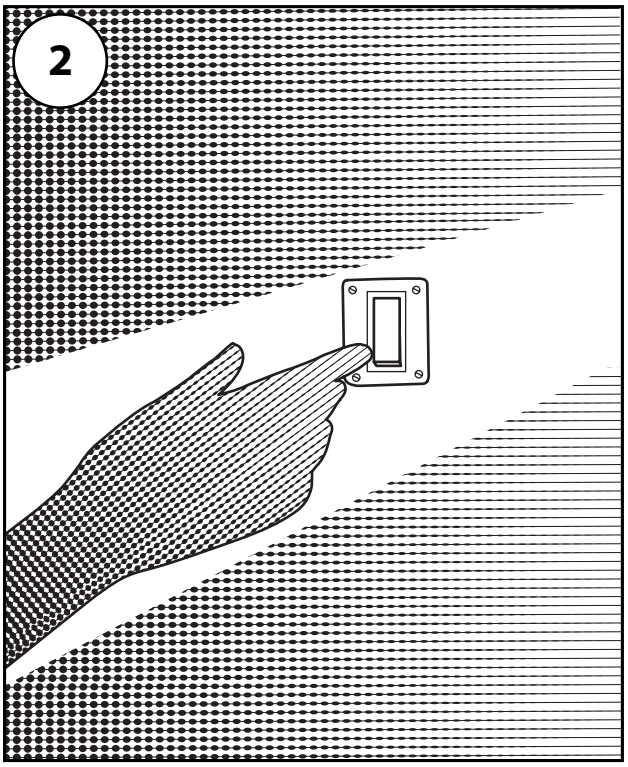
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Activity Page 2.1 Use with Lesson 2.

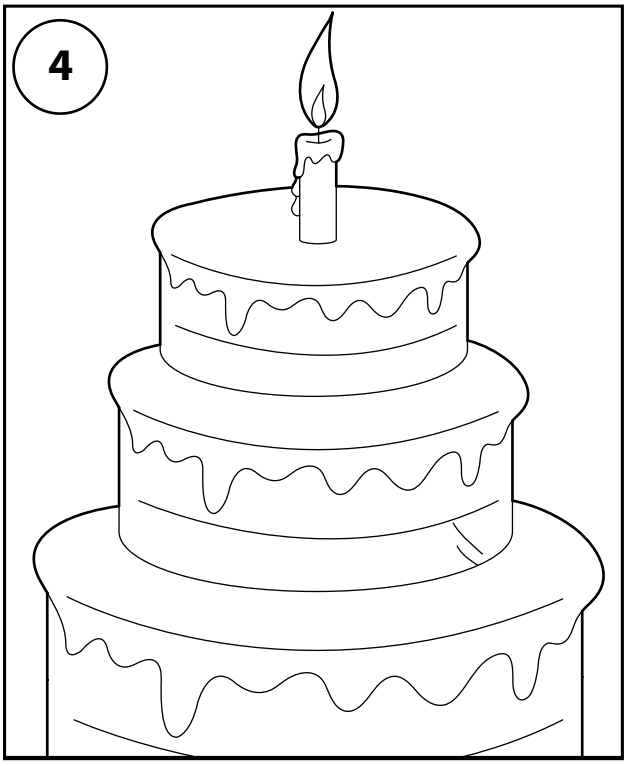
Surprise!

What changes occur in the party pictured here?









Name _____ Date _____

Activity Page 2.2 Use with Lesson 2.

Energy Causes Change

light	sound	electrical energy	motion energy	stored energy

<u>Choose</u> one form of energy from the box above. Circle your choice.

What change can this form of energy cause?

_____

_____

<u>Draw</u> a "before and after" cartoon to show a change, and indicate the form of energy that causes the change.

Before	After

<u>Explain</u> the cause and effect shown in your "before and after" cartoon.

What causes the change? What changes? (In other words, what is the cause, and what is the effect?)

_____

_____

_____

Name _____ Date _____

Activity Page 3.1 Use with Lesson 3.

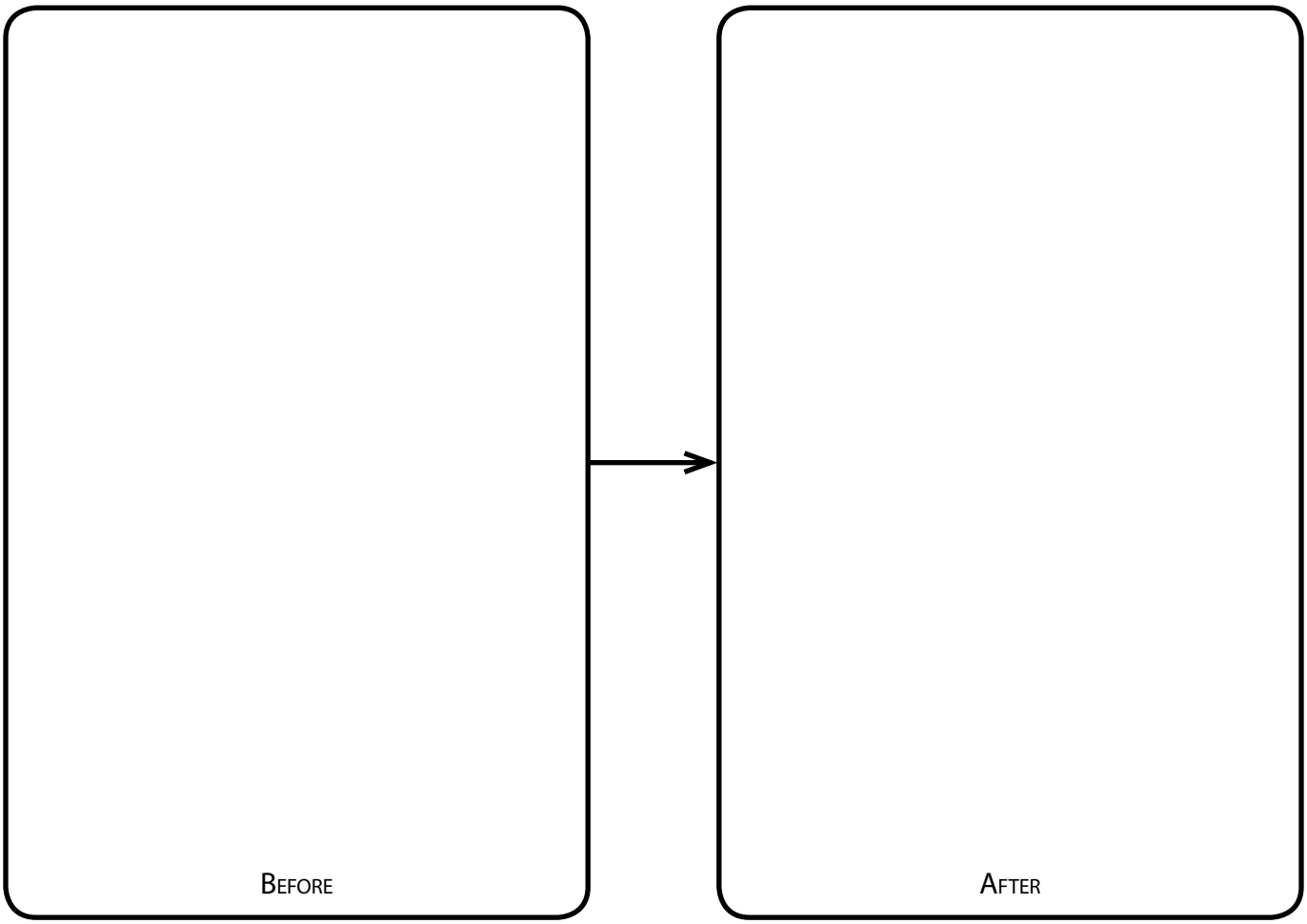
On the Move

Things are on the move everywhere that you look. Those moving objects can cause changes.

Identify an object that you know moves often.

_____

Draw before and after diagrams to show how the motion of this object can cause a change.



Write descriptions to explain your model.

_____ _____

_____ _____

_____ _____

Name _____ Date _____

Activity Page 4.1 Use with Lesson 4.

Ramp It Up (Day 1)

Different objects are moving at different **speeds** all around you! From the cars on the road to balls on the playground, all moving objects have speed and energy.

Is there a connection between speed and energy? You are about to find out!

STEP 1: Build a low ramp from cardboard propped at one end by a book. Place the cup a short distance from the bottom of the ramp (about the length of your foot).

STEP 2: Trial 1 Place your ball at the top of the ramp. Release the ball so it rolls down the ramp into the cup. Notice how far the ball causes the cup to move. Place a marker to show how far the cup moved.

STEP 3: Add books to make your ramp steeper so your ball will roll from a greater height. Place the cup in the same position in front of the ramp as you did before.

STEP 4: Trial 2 Place your ball at the top of the ramp. Release the ball so it rolls down the ramp into the cup. Notice how far the ball causes the cup to move.

STEP 5: Use your observations to complete the table.

	Trial 1	**Trial 2**
In which trial was the ramp height **higher**? In which trial was it **lower**?		
In which trial did the ball roll with **greater speed**? In which trial did it roll with **lesser speed**?		
In which trial did the ball move the cup **more**? In which trial did it move **less**?		
In which trial did the rolling ball have **more energy**? In which trial did it have **less energy**?		

STEP 6: Summarize your conclusion. Describe the relationship between energy and speed in your investigation. Use evidence to support your explanation.

Name _____ Date _____

Activity Page 4.2 Use with Lesson 4.

Drop It (Day 2)

Dropping an object from different heights produces different falling speeds for the object.

Does the relationship between speed and energy remain the same as what you observed when rolling a ball down a ramp?

STEP 1: Smooth the sand in the box so that there are no dents in the surface of the sand.

STEP 2: Trial 1 Hold the ball above the sand near one end of the box (not in the middle). Hold the ball at about knee-height.

STEP 3: Drop the ball into the sand. Gently pick the ball out of the sand without disrupting the sand. Notice the size of the crater that the ball's impact made in the sand.

STEP 4: Trial 2 Hold the ball above the sand near the opposite end of the box. Hold the ball at about waist-height.

STEP 5: Drop the ball into the sand again. Gently pick the ball out of the sand without disrupting the sand. Notice the size of the crater that the ball's impact made in the sand this time.

STEP 6: Use your observations to complete the table.

	Trial 1	**Trial 2**
In which trial was the ball dropped from **higher** position? In which trial was it dropped from a **lower** position?		
In which trial did the ball drop with **greater speed**? In which trial did it drop with **lesser speed**?		
In which trial did the ball have a **greater impact** on the sand? In which trial did it have **less** impact?		
In which trial did the falling ball have **more energy**? In which trial did it have **less energy**?		

STEP 7: Summarize your conclusion. Describe the relationship between energy and speed in your investigation. Use evidence to support your explanation.

Name _____ Date _____

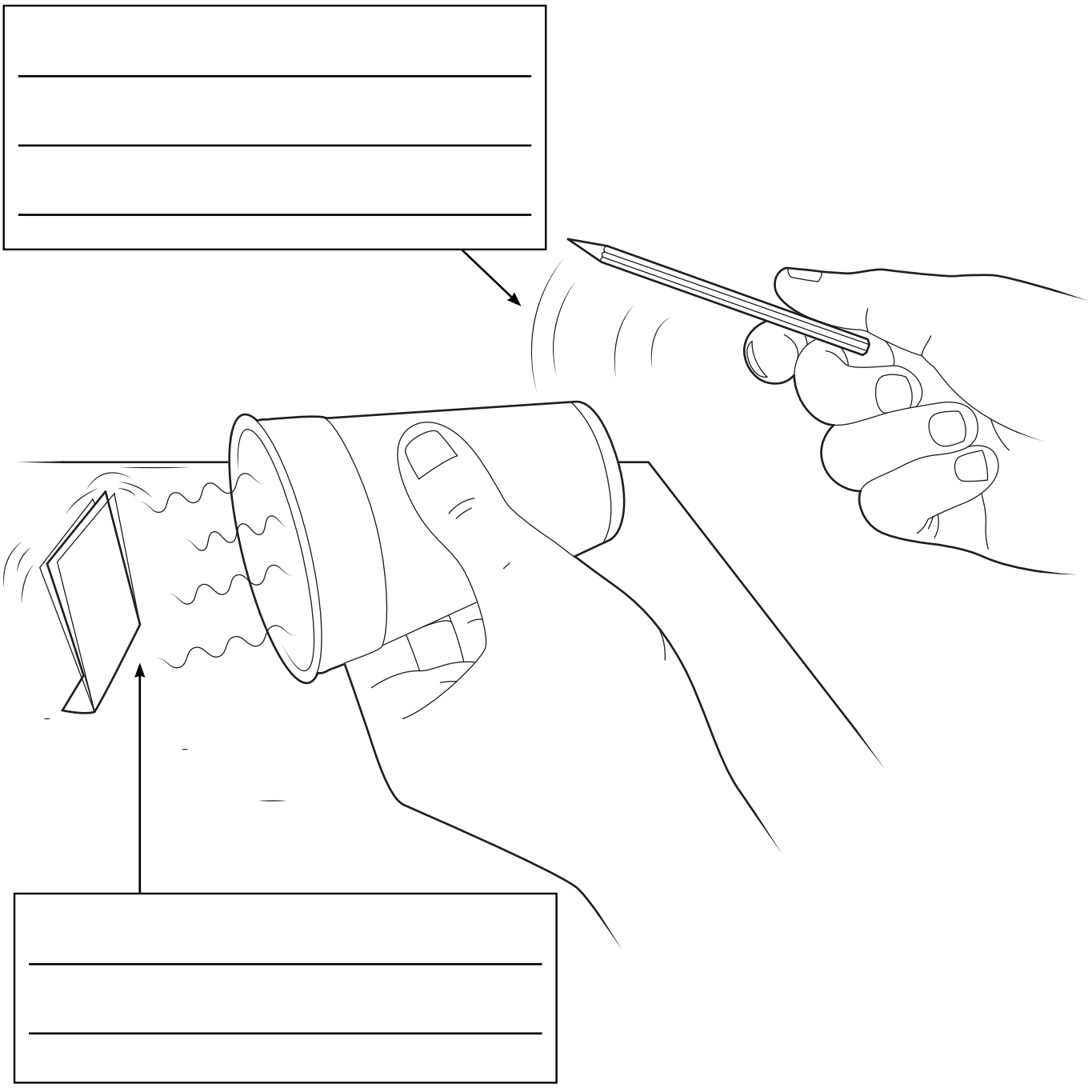
Activity Page 6.1 Use with Lesson 6.

Lesson 6 Check

<u>Write</u> captions on the picture to explain what is happening.

Use each of these words at least one time in your captions.

sound	energy	evidence	transfer



Name _____ Date _____

Activity Page 7.1 Use with Lesson 7.

Lesson 7 Check

Answer the items below to show what you have learned.

1. Which of the following is **evidence** of energy transfer? Circle all that apply.

a) a ball sitting at the top of a ramp

b) the sound from a vibrating tuning fork

c) a lighted wall across the room from a lamp

d) a crater produced by a meteorite impact

2. Draw a picture showing **energy transfer** from one object to another object by heat or electricity. Write a caption on the lines below to explain your drawing. Use the word ***evidence*** in your caption.

Name _____ Date _____

Investigating Energy Transfer

Station 1: Light Transfers Energy

STEP 1: Turn the flashlight on, and shine it at the wall so that its beam passes above the plate of chalk powder.

STEP 2: Gently dip the paintbrush bristles into the plate of chalk powder. Be careful to not make a mess!

STEP 3: Tap the brush gently with your free hand so the chalk powder falls back onto the plate. Watch as the chalk falls between the flashlight and the wall.

STEP 4: Describe what you see happen in terms of motion energy, chemical energy, electrical energy, and light energy.

STEP 5: Leave the station ready for the next group.

- Turn the flashlight off.
- Place the paintbrush back on the table.
- Be sure to let your teacher know if there is not enough chalk powder left on the plate. Other groups will need to repeat the same activity after you!

STEP 6: Answer the questions below about what you observed. Repeat the investigation if you need to.

1. Can you see a beam of light between the flashlight and the wall without the chalk dust?

2. How does the chalk dust change what you are able to observe?

3. What evidence did you observe that energy transferred from place to place?

Station 2: Heat Transfers Energy

SAFETY NOTE: Do not touch the hot plate or the beaker.

STEP 1: Use the tongs to place four ice cubes from the icebox/cooler into the beaker.

STEP 2: Watch what happens to the ice cubes.

STEP 3: Describe what you see happen in terms of electrical energy and heat energy.

STEP 4: When you are finished observing, let your teacher know to ready the station for the next group.

STEP 5: Answer the questions below about what you observed. Repeat the investigation if you need to.

1. What would happen if the temperature of the hot plate was decreased? How might this change the result?

2. Would the results change if you added more ice cubes? Why or why not?

3. What evidence did you observe that energy transferred from place to place at this station?

Station 3: Electricity Transfers Energy

SAFETY NOTE: Do not handle the surge protector power cord that is plugged into the wall electrical socket.

STEP 1: Observe the electronic device sitting on the table. Unplug the device from its charging cord. Notice what happens. Plug the charging cord back into the device. Notice what happens.

STEP 2: Trace the cord all the way to the surge protector. Switch the surge protector into the *off* position. Study the electronic device. Switch the surge protector back to the *on* position. Notice what changes.

STEP 3: Describe what you see happen in terms of electrical energy and stored energy.

STEP 4: Prepare the station for the next group.

- Plug the device back in (if it is unplugged).
- Turn the surge protector to the *on* position.
- Place the device on the table.

STEP 5: Answer the questions below, and repeat the investigation if you need to.

1. What causes a change in the device?

2. What happens when you turn the surge protector off? What happens when you turn it on?

3. What would happen if there were not a wire connecting the surge protector to the device?

4. What evidence did you observe that energy transferred from place to place at this station?

Name _____ Date _____

Activity Page 9.1 Use with Lesson 9.

Examples of Energy Transfer

You can observe examples of energy being transferred every day.

Identify and record some real-life examples in the table below.

Type of Energy	Description of Energy Transfer
Sound	
Light	
Heat	
Electricity	

Name _____ Date _____

Activity Page 9.2 (Page 1 of 2) Use with Lesson 9.

Claims, Evidence, and Reasoning About Energy Transfer

Part 1

Let's take a moment to review the important terms **claims**, **evidence**, and **reasoning**:

1. Claim:

What is a **claim**?

_____

_____

Why do people make **claims**?

_____

_____

2. Evidence:

What is **evidence**?

_____

_____

What are some types of **evidence**?

_____

_____

3. Reasoning:

Describe an example of scientific **reasoning**.

_____

_____

Why do you think people use **reasoning** in science?

_____

_____

Part 2

Now it's time to write a claim of your own.

1. Choose one example of energy transfer that you have learned about or investigated. Describe an example of how this form of energy transfers from one place to another.

2. What kind of evidence (data, observations, facts) was there that energy transfer took place?

3. Write a scientific claim about energy transfer. Follow this model: "I know _____ because_____ ."

Name _____ Date _____

Activity Page 10.1 Use with Lesson 10.

Lesson 10 Check





1. What two objects are about to collide in the picture above?

_____

_____

2. What form of energy can be observed before the collision happens?

_____

3. What forms of energy would you expect to observe as a result of this collision? Support your answer with evidence.

_____

_____

_____

_____

4. What two objects are about to collide in the picture above?

_____

_____

5. What form of energy can be observed before the collision happens?

_____

6. What forms of energy would you expect to observe as a result of this collision? Support your answer with evidence.

_____

_____

_____

_____

Name _____ Date _____

Activity Page 11.1 Use with Lesson 11.

Investigating Collisions (Day 1)

Predictions

Plan your investigation by completing the items below.

1. What do you want to find out? (Example: I want to find out what happens to energy when objects collide.)

2. What materials will you use for your investigation?

3. How will you make them collide?

4. Make a prediction. What do you think will happen when the objects collide?

5. What evidence might you observe that indicates energy transfers from one object to another?

Name _____ Date _____

Activity Page 11.2 (Page 1 of 2) Use with Lesson 11.

Investigating Collisions (Day 2)

Testing and Observation

Complete your investigation by completing the items below.

1. Use the materials you chose on Day 1, and build your ramp.

2. Conduct the investigation.

- Place the objects on the ramp.
- Roll the top objects down the ramp, and observe what happens.

3. Record your observations in the table below under Test 1.

Test	**Observations of Energy Transfer**
Test 1	
Test 2	

4. Change one variable (i.e., height or speed) in your design. Which variable will you change?

_____

5. Make a new prediction. What do you expect to see now that you changed this variable?

_____

_____

_____

6. Build your second design.

7. Perform the test again with the changed variable.

8. Record the new observations for Test 2 in the table.

9. Study the observations in the table. Answer the following questions:

- Was any energy transferred? Use evidence to support your answer.

- What are the differences between the two tests?

- Do you see any patterns? If so, what are they?

- What is the cause-and-effect relationship that you see? Write your answer as an *If, Then* statement.

  If _____ ,

  then _____

10. Evaluate the predictions you made before each test. Were your predictions accurate? Why or why not?

Name _____ Date _____

Activity Page 12.1 Use with Lesson 12.

Lesson 12 Check

Answer the questions to show what you know from this lesson.

1. What is the Big Question for this lesson?

_____

_____

2. Name a device that converts stored chemical energy of natural gas to heat energy to help people cook food.

_____

3. What energy transformation(s) takes place in a smoothie blender?

_____

_____

4. What problem does a wind-up kitchen timer solve? Use the word *solution,* as well as *want* or *need* (or both), in your answer.

_____

5. Describe two energy transformations that happen in a wind-up kitchen timer.

_____

_____

_____

6. How do people use a conversion from electrical energy to light energy in helpful ways in a kitchen or school classroom?

_____

_____

Name _____ Date _____

Activity Page 13.1 (Page 1 of 2) Use with Lesson 13.

Device Design Proposal (Day 1)

<u>Plan</u> a device to solve a problem using an energy transformation.

Problem to Be Solved: _____

_____

Device Name: _____

Energy Transformation:

Initial Form(s) of Energy	Final Form(s) of Energy
_____	_____
_____	_____

How will your team's device solve the problem by transforming (converting) energy?

_____

_____

_____

How will you know whether your device solves the problem or not?

_____

_____

Criteria: _____

Constraints (limitations): _____

Materials: Circle the materials your team will use. Write in any additional materials used on the lines below.

aluminum foil	cardboard tubes	cups	springs
ball	clear plastic wrap	glue	sticks
blocks	containers	ramps	string
boxes	craft paper	rubber bands	tape

_____

Name _____ Date _____

Activity Page 13.1 (Page 2 of 2) Use with Lesson 13.

Draw your device, and add labels to help show how your device will work.

Name _____ Date _____

Activity Page 13.2 Use with Lesson 13.

Device Test Results (Day 2)

Trial Test your device multiple times.	**Results** Briefly describe whether your device worked or not.	**Observations About Energy Transfer and Transformation** What evidence do you have that your device worked?	**Improvements or Fixes** How can you refine your device to work better in the future?
1.			
2.			
3.			

Make a claim about whether testing your device is important or not. Use your observations as evidence and reasoning to support your claim.

Name _____ Date _____

Activity Page 13.3 Use with Lesson 13.

Device Presentation Plan (Day 3)

Review your work on Activity Page 13.2 (Day 2). Summarize your work so far by answering these questions.

Problem to Be Solved: _____

_____

Device Name: _____

Energy Transformation:

Initial Form(s) of Energy	Final Form(s) of Energy
_____	_____
_____	_____

How well did your device solve the problem? _____

_____

_____

How did you build your device? _____

_____

_____

What were the results of your tests? _____

_____

_____

How did you refine your device? _____

_____

_____

Describe whether your revisions made your device work better or worse. Support your ideas by summarizing observations and evidence from each of your test trials.

_____

_____

Name _____ Date _____

Activity Page 13.4 Use with Lesson 13.

Device Presentation Scoring Guide (Day 4)

Use this guide as you listen to your classmates present their devices. For each line of the guide, give the team a score that reflects how well their presentation covered that section. For example, a score of 3 for "Defines the problem to be solved" would mean the presentation clearly told others of the problem to be solved.

Description	3	2	1
Defines the problem to be solved			
Meets the criteria defined for the device			
Meets the constraints or limitations defined for the device			
Identifies the initial and final forms of energy in the energy transformation			
Describes how the solution converts energy from one form to another to solve the problem			

I like your group's design because _____

_____

_____

What other questions would you ask about the device?

_____

_____

_____

Name _____ Date _____

Energy Vocabulary Crossword Puzzle

Use the words in the word bank to complete the crossword puzzle. Review the cards in your Core Vocabulary deck before you begin.

light	change	evidence	energy	motion	forms of energy*	position
sound	convert	heat	stored energy*	energy transfer*	electricity	
speed	energy of motion*	distance				

*Spaces that appear between words in terms above do not appear in the puzzle.

Across

2. when energy moves from one object to another or from one place to another

4. a type of energy that makes things hot

8. different types of energy, such as sound, heat, or light

11. the energy an object has while it is moving

12. to change

13. form of energy that powers lights in modern homes

14. how fast an object is moving

15. anything that helps prove or disprove an idea

16. to become different

Down

1. where something is in space

3. the process of an object changing position

5. a type of energy you can see with your eyes

6. the ability to cause change

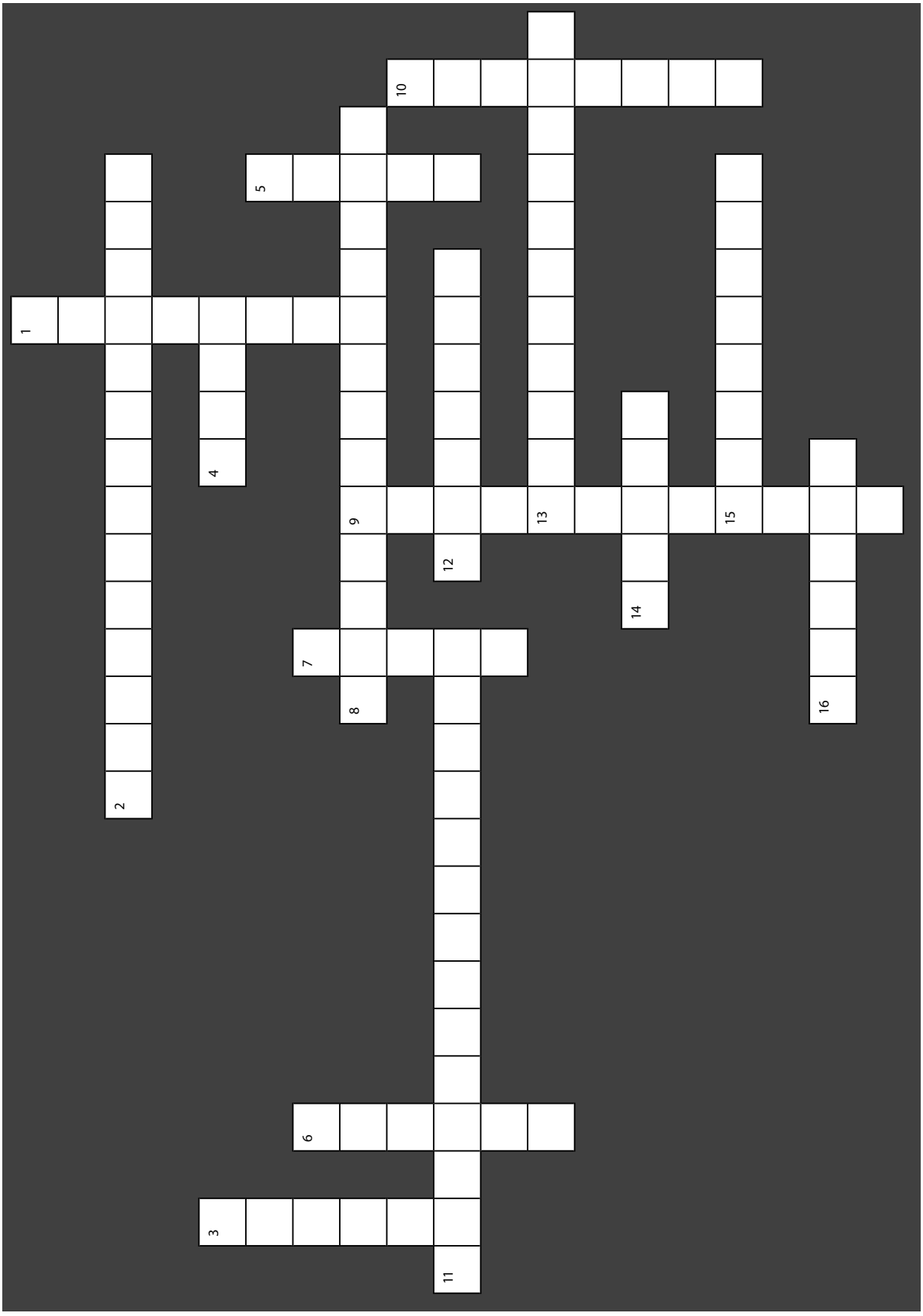
7. a type of energy you can hear with your ears

9. energy that has the potential to cause change at a later time

10. how far apart objects are

Name _____ Date _____

Energy Vocabulary Crossword Puzzle



Name _____ Date _____

Activity Page 13.6 Use with Lesson 13.

Energy Vocabulary Review

Complete each sentence with the correct Core Vocabulary term or phrase. Not every word in the word bank will be used, and some may be used more than once. Review the cards in your Core Vocabulary deck before you begin.

change	energy	forms of energy	stored energy	evidence	heat	light
motion	position	sound	transfer	collision	electricity	observation
model	speed	distance	energy of motion	convert	energy transformation	

1. When you add hot soup to a cold bowl, _____ energy transfers from the soup to the bowl.

2. If a book is falling off a shelf, the book has _____ because it is changing its _____.

3. Playing a drum produces _____, which is a form of energy that you can hear.

4. Energy can _____ from one object to another object.

5. When a bowling ball hits the bowling pins, a sound rings through the bowling alley. The change from motion energy to sound energy is an example of a(n) _____.

6. Energy is the ability to cause _____.

7. Heat, light, and sound are examples of _____.

8. A marble at the top of a ramp has _____ , because at any moment it can move down the ramp.

9. Computers are powered by _____.

10. _____ is a measure of how fast an object is moving, and _____ is a measure of how far apart objects are.

11. When light energy is converted to heat energy, this is known as _____.

Name _____ Date _____

Activity Page UR.1 Use with Unit Review.

Energy's Big Questions

The table below contains most of the Big Questions you have explored in the unit *Energy Transfer and Transformation*.

Circle the question that was assigned for you to answer.

Where can we observe evidence of energy causing change?	What evidence shows that energy is transferred from place to place?
What are some forms of energy?	How is energy involved in collisions?
How are energy, change, and movement of objects related?	What happens when objects collide?
How are energy and speed related?	How are collisions predictable?
What evidence shows that sound transfers energy from one place to another?	How do energy transformations help people?
What evidence shows that light, heat, and electricity transfer energy from place to place?	How did Thomas Edison use his knowledge of energy transfer and transformation to solve problems?

Write three or four sentences to answer the Big Question that was assigned to you and your partner.

Prepare a rough draft of your answer first on scrap paper, and then write your final draft below. Underline any terms in your answer which have a card in your Core Vocabulary deck.

Name _____ Date _____

Unit Assessment: What Have I Learned About Energy?

Answer the items below to show what you have learned.

1. Which sentences describe evidence of energy causing change? Circle the letter for each correct answer.

a) A basketball smashes into a backboard.

b) Electricity jumps from your fingertip to a doorknob, giving you a shock.

c) When the school bus starts up, you can hear the engine make a sound.

d) A stick of dynamite explodes.

2. Write the form or forms of energy described in each sentence below. Some sentences may describe more than one form of energy.

light	heat	motion	sound	electrical

a) The sun is shining on a nice summer day. _____

b) A firework explodes in the sky. _____

c) You kick a soccer ball at the goal. _____

d) You're happy and you know it, so you clap your hands. _____

e) You buckle your seat belt, and the school bus takes off to the school. _____

f) You turn on a battery-powered flashlight. _____

3. Draw a model of energy causing a change. Your drawing may show a before-and-after transformation, or you can use your imagination. Identify the type of energy causing the change. Write a sentence describing the change.

Type of energy: _____

Describe the change: _____

_____

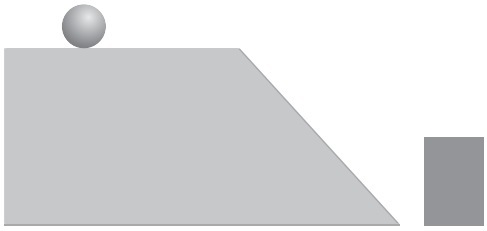
4. You are setting up an investigation to show the relationship between the speed of a moving object and its energy. You have a ball and a block. You also have three different types of ramps: a low-height ramp, a medium-height ramp, and a high-height ramp.



low



medium



high

Which ramp will result in the greatest amount of change? Explain your thinking using words from your Core Vocabulary deck.

5. Which ramp will result in the least amount of change? Explain why using words from your Core Vocabulary deck.

6. What is the relationship between the speed of an object and its energy of motion? Use evidence and examples you have learned during this unit to support your explanation.

7. Maria has been studying sound energy at school. She has written an explanation of sound using an example from her experience.

Maria says, "Sound energy only travels over long distances. When I am on the playground, I can hear the bell that rings from across the field that lets me know recess is over."

Do you agree with Maria? Use your own evidence to support your answer.

8. Which statements are evidence that energy has been transferred from object to object or place to place? Circle the letter for each correct answer.

a) Your friend kicks a goal while playing soccer.

b) Gravity pulls on an apple, but it doesn't fall out of a tree.

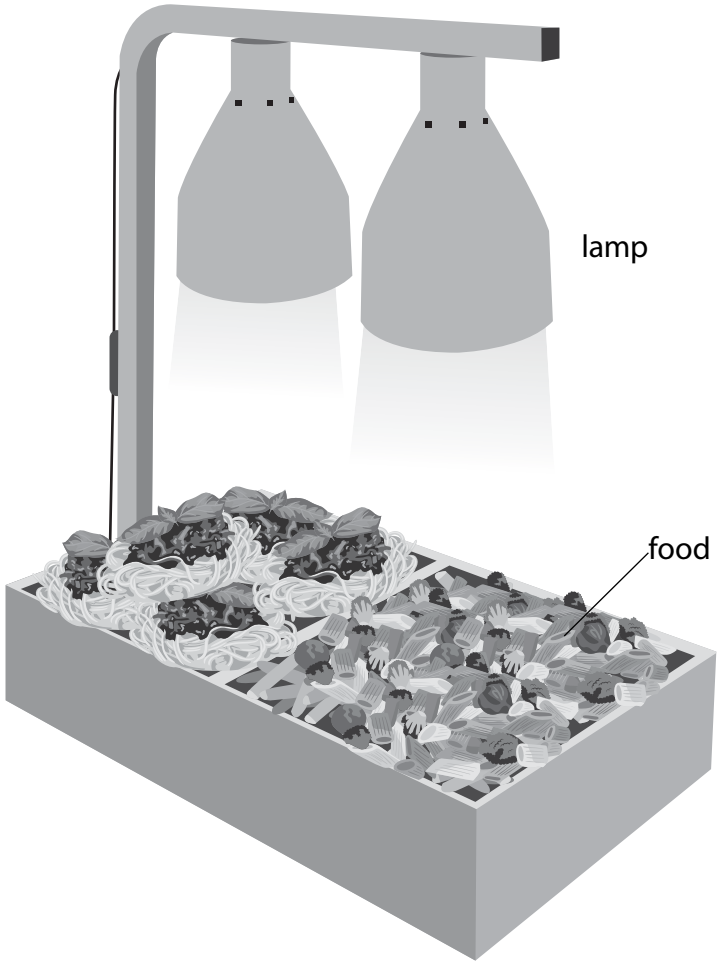
c) You push on the school door, but it stays closed.

d) You hear a bell ringing from far away.

e) You flip a light switch, and the room lights up.

f) A person pulls a wagon down a street.

g) The backpack you left in the car gets hot on a sunny day.



The school cafeteria keeps cooked food under lamps until students pick it up to eat. The top of each portion of food stays warmest.

9. What two kinds of energy are the lamps transferring to the food?

10. What evidence would support your claims about the types of energy used in the cafeteria example above? Use terms from your Core Vocabulary deck in your answer.

11. You roll a toy car down a ramp into a stack of four blocks. The top block falls off the stack.

You add some weight to the toy car and repeat the test, changing ONLY the weight of the car. This time, the top two blocks fall off the stack.

Based on this pattern, explain the result in terms of energy change.

What can you predict if even more weight is added to the toy car for the next test? Use terms from your Core Vocabulary deck in your answer.

12. Your teacher holds a ball up and drops it. She asks what happened to the energy of the ball. Here are the answers from four students.

Student A: The stored energy of the ball becomes motion energy.

Student B: The chemical energy of the ball becomes motion energy.

Student C: Heat transforms into motion energy.

Student D: Gravity is transferred into heat energy.

Which student is correct? Explain your reasoning using vocabulary, such as *position*, and examples that you have learned during this unit.

13. Your teacher is leaving over the weekend, but the house plants in his classroom can't be left without water or light. Describe a device that could solve this problem. The solution would need to automatically provide water **and** light to a plant when people are away.

14. If you were to build your device for your teacher, how would you test it to see if it works, and how might you improve your design based on your test results?

15. A student named Maria made a sketch of her solution to the same problem. Maria's solution includes plugging a lamp into a timer that works in a wall socket. At the same times each day, the timer switches the lamp on and off. Her solution also includes a container of water that allows water to drip slowly through a hose down into the planter. The container of water sits on a stack of books so it is positioned higher than the planter.

Complete the labels on Maria's diagram to identify the types of energy that are converted in her design.

Maria's drip hose solution converts

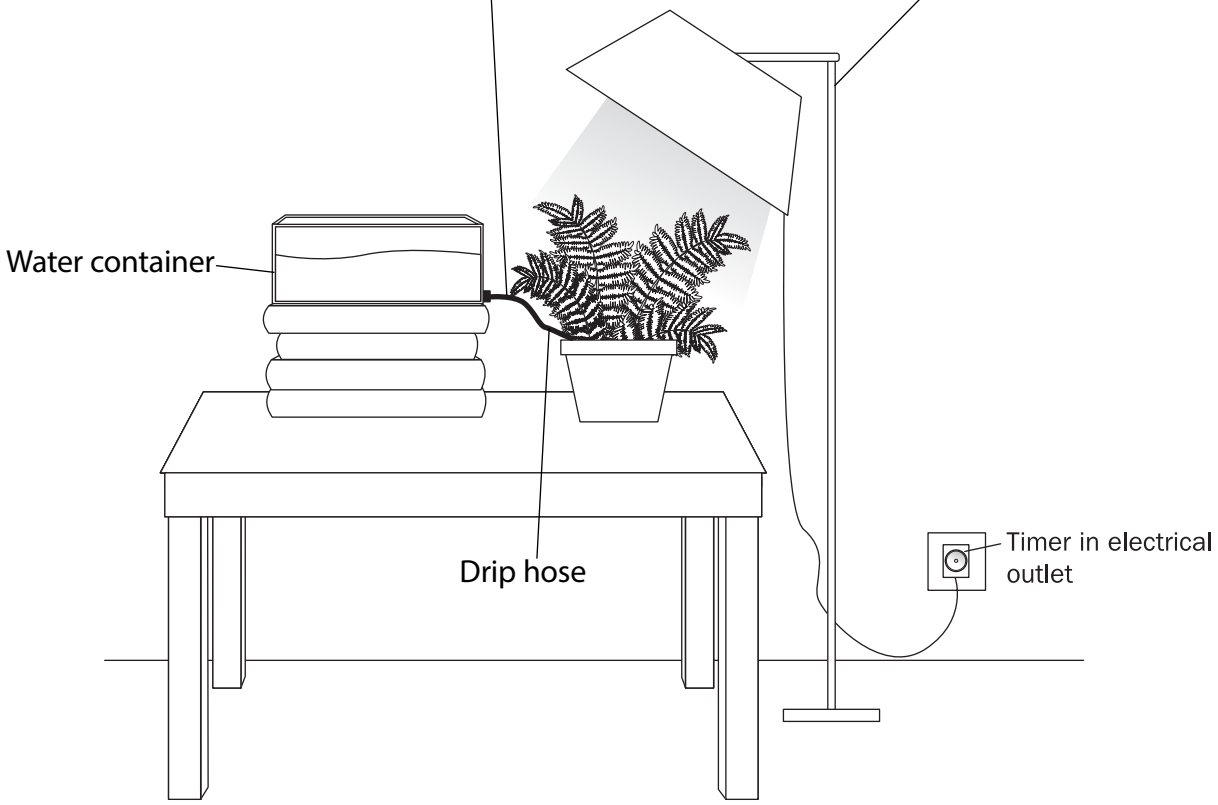
_____

to _____ .

Maria's lamp design solution converts

_____

to _____ .



Activity Pages Answer Key: *Energy Transfer and Transformation*

This answer key offers guidance to help you assess your students' learning progress. Here, you will find descriptions of the expectations and correct answers for each of the Activity Pages of this unit.

Energy Scavenger Hunt (AP 1.1) (page 101)

- Students should include examples of three changes and a brief explanation of how each change occurs.
- Accept any reasonable student questions that are based on the examples the students wrote.

Surprise! (AP 2.1) (page 102)

1. Light energy transfers (moves) into the room.
2. Electrical energy transforms into light energy as the light is turned on.
3. Sound energy is made by the people talking and the noisemakers.
4. Light and heat energy are released by the candle.

Energy Causes Change (AP 2.2) (page 103)

- Students should correctly identify a change caused by the energy they circled in the word box, such as stored energy changing to motion energy when a person walks across the room.
- Before and after drawings should be correct and include the form of energy causing the change, such as drawing the candle without a flame and then a lit candle.
- Students should identify both the cause and the effect. For example, a match can be struck to light the candle, and the effect is light and heat.

On the Move (AP 3.1) (page 104)

- Students should identify an object that moves.
- Before and after drawings should be correct and show the change.
- Students' descriptions should include correct before and after descriptions based on their drawings.

Ramp It Up! (Day 1) (AP 4.1) (page 105)

Step 5 Students should complete the table for each trial. The first trial should show a lower starting height, less speed, less cup movement, and less energy. The second trial should show a higher starting height, greater speed, more cup movement, and more energy.

Step 6 Students should use their results to note the relationship between energy and speed, citing evidence to support their explanation. A steeper ramp makes the ball roll faster. The greater speed has more energy when it hits the cup, so it moves farther.

Drop It (Day 2) (AP 4.2) (page 106)

Step 6 Students should complete the table for each trial. The first trial should show a lower starting height, less speed, less impact, and less motion energy. The second trial should show a higher starting height, more speed, greater impact, and more motion energy.

Step 7 Students should use their results to describe the relationship between energy and speed, citing evidence to support their explanation. A drop from a greater height makes the ball drop faster. The greater speed has more energy when it hits, so the impact makes a bigger crater.

Lesson 6 Check (AP 6.1) (page 107)

Students should note that the pencil transfers energy of motion to the bottom of the cup and that the sound of the tapping is evidence of energy transfer. Students should also note that sound from the tapping comes out of the top of the cup and that the movement of the paper is evidence of energy transfer.

Lesson 7 Check (AP 7.1) (page 108)

1. b, c, d
2. Student drawing should show a clearly captioned and explained energy transfer by heat or electricity. The term *evidence* is used in the caption.

Investigating Energy Transfer – Station 1: Light Transfers Energy (AP 8.1) (page 109)

Step 4 Students should note that light energy is transferring from the flashlight to the wall.

1. Students should note that the flashlight batteries transfer chemical energy to light.
2. Students should note that they cannot see the beam of light without the chalk dust.
3. Students should note the chalk dust gives the light energy something to bounce off.
4. Students should note the evidence of energy transfer is the light shining on the chalk dust and wall.

Investigating Energy Transfer – Station 2: Heat Transfers Energy (AP 8.1) (page 110)

Step 3 Students should note that electrical energy is transferring to the hot plate. Then the heat energy is transferring from the hot plate to the beaker and then to the ice cubes.

1. Students should note that less heat energy would flow from the hot plate to the beaker, causing the ice cubes to melt slower.
2. Students should note that more ice cubes would take longer to melt because more heat energy would be needed.
3. Students should note the melting of the ice cubes is evidence that energy is being transferred from place to place.

Investigating Energy Transfer – Station 3: Electricity Transfers Energy (AP 8.1) (page 111)

Step 3 Students should note the electrical energy is transferring from the outlet to the surge protector and then to the device.

Station 3, continued

1. Students should note that the electrical energy flowing from the surge protector to the device causes a change.
2. Students should note that when the surge protector is off, no electrical energy flows and that when it is on, electrical energy flows.
3. Students should note that the electrical energy would not flow if there were no wire.
4. Students should note the device is signaling when electrical energy is flowing into the device.

Examples of Energy Transfer (AP 9.1) (page 112)

Students should include examples from all four types of energy listed and should include an accurate description of the transfer.

Claims, Evidence, and Reasoning About Energy Transfer (AP 9.2) (pages 113–114)

Part 1

1. Claim: A claim is a statement that declares a fact or presents an argument that has not been proven. People make claims as a key step to making a scientific examination.
2. Evidence: Evidence is proof that supports or refutes a claim. An example of evidence would be the results from a science investigation.
3. Reasoning: Student examples of reasoning should show a clear and logical progression. People use reasoning in science because they seek to reach testable truths.

Part 2

1. Student examples should indicate a clear transfer of energy, such as hot soup being poured into a bowl and subsequent temperature changes.
2. Student evidence should refer back to their learning or investigation, such as noting that the bowl that the hot soup is poured into gets hot, indicating the transfer of heat energy.
3. Student responses should be properly supported and in the form of "I know . . . because . . .," such as, "I know the heat energy in the soup transferred to the bowl because the bowl is now hot."

Lesson 10 Check (AP 10.1) (page 115)

1. The soccer player's foot and the soccer ball are about to collide.
2. The form of energy before the collision is energy of motion.
3. There would be energy of motion, which you could see as the ball moves. There would be sound energy that you could hear. There would be a tiny amount of heat energy from the collision.
4. The drumsticks and drum are about to collide.
5. The form of energy before the collision is energy of motion.
6. There would be energy of motion as the drumsticks bounce back up. There would be sound energy, which you could hear. There would be a tiny amount of heat energy from the collision.

Investigating Collisions (Day 1): Predictions (AP 11.1) (page 116)

Sample responses:

1. We want to find out how energy causes changes during a collision and if the amount of energy changes with speed.
2. We will use a ramp and send a toy car down the ramp. It will collide with a can, and we can see how far the can moves.
3. The toy car will collide with the can.
4. I predict the toy car will push the can back.
5. We will roll the car down the ramp to observe it. We will measure the distance the can moves by using car lengths.

Investigating Collisions (Day 2): Testing and Observation (AP 11.2) (pages 117–118)

3. Student observations should show that the object travelling down the ramp collided with and moved the object at the bottom of the ramp.
4. Students should identify one independent variable to change, such as the height of the ramp or the weight of an object.
5. If the toy car has more weight, it will move the can farther back when they collide.

Investigating Collisions (Day 2), continued

9.
   - Energy of motion was transferred from the toy car to the can.
   - Student observations between the two tests should be consistent with the change in variables, e.g., adding more weight to the car or height to the ramp caused a collision with more energy, resulting in the effect of the can moving farther.
   - The more energy added before the collision, the more energy we saw during the collision.
   - If more energy is added to a collision (cause), then we will see more movement (effect) after the collision.
10. We predicted adding more weight to the car would result in more energy in the collision. The prediction was accurate as the can moved farther after the collision in the second test.

Lesson 12 Check (AP 12.1) (page 119)

1. How do energy transformations help people?
2. an oven or stove
3. Electrical energy is converted to energy of motion and sound energy.
4. A kitchen timer provides the solution of what to do when you need to cook something a specific length of time so it does not overcook or undercook.
5. The stored energy of the spring inside the timer changes to energy of motion. When the bell rings, energy of motion changes to sound energy.
6. Accept all reasonable responses, such as, "The lights in the classroom allow the change of electrical energy into light energy so students can read their books."

Device Design Proposal (Day 1) (AP 13.1) (pages 120–121)

- The student problem to be solved should be their assigned problem.
- Students should identify the initial and final forms of energy their device will transform.
- Student descriptions of their device should include how the device solves the problem and some reference to how the energy will be transformed.
- Accept all plausible criteria and constraints.

(Day 1), continued

- Students should circle or list any materials they will use in their device.
- Student drawings of their device should include clear incorporation of their materials.

Device Test Results (Day 2) (AP 13.2) (page 122)

- Student results and observations may vary, but there should be improvements or fixes to their devices based on the test results.
- Claims should indicate that testing their device was an important step in improving the device.

Device Presentation Plan (Day 3) (AP 13.3) (page 123)

- Students should identify the problem, criteria, and energy transformation from Device Design Proposal (AP 13.1).
- Students should provide a clear description of how they built their device.
- Students should refer to Device Test Results (AP 13.2) for initial information about their test results, which they can then expand on.
- Students should refer to Device Test Results (AP 13.2) for initial information about their refinements to their device, which they can then expand on.

Device Presentation Scoring Guide (Day 4) (AP 13.4) (page 124)

Students should complete the scoring guide for the device. Students should also identify what they liked in the other group's device, as well as other questions they may have had about the device.

Energy Vocabulary Crossword Puzzle (AP 13.5) (pages 125–126)

ACROSS:

2. energy transfer

4. heat

8. forms of energy

11. energy of motion

12. convert

13. electricity

14. speed

15. evidence

16. change

DOWN:

1. position

3. motion

5. light

6. energy

(AP 13.5), continued

7. sound

9. stored energy

10. distance

Energy Vocabulary Review (AP 13.6) (page 127)

1. heat **2.** energy of motion, position **3.** sound **4.** transfer **5.** energy transformation **6.** change **7.** forms of energy **8.** stored energy **9.** electricity **10** speed, distance **11.** energy transformation

Energy's Big Questions (AP UR.1) (page 128)

Students should circle their assigned question. Student responses should reflect the content and experiences from the lesson(s) from which the Big Question originates. Vocabulary terms from their Core Vocabulary decks should be incorporated in their written responses and underlined.

Consider student responses as follows:

Above Average—Written responses clearly identify examples and ideas that help answer the Big Question. Examples are clearly linked to the Disciplinary Core Ideas learned, and Core Vocabulary is incorporated into the response and underlined.

Average—Written responses may include examples of learning about the Big Question but do not clearly link to the DCIs learned during the lessons. Core Vocabulary is incorporated into the response.

Adequate—The Big Question is answered, but not using examples from the lessons or Student Reader chapters. Core Vocabulary may be listed but is not used in a complete sentence, and/or a minor misunderstanding about the Disciplinary Core Ideas is present.

Inadequate—Responses do not relate to the Big Question, or responses include a clear misunderstanding of the Big Question or DCI.

Unit Assessment: Teacher Evaluation Guide

Teacher Directions: The Unit Assessment on pages 129–136 is designed as a fifty-point test. Through this assessment, students demonstrate their overall learning of the unit's Learning Objectives. CKSci Unit Assessments typically range from ten to fifteen questions in the upper elementary grades, which can be answered in a longer, single classroom session or administered in two sittings.

Items with simpler answers that assess knowledge but not the deeper understandings of the content, such as multiple choice or short answers, are weighted differently and are worth fewer points. Assessment items that require more complex thinking and a deeper understanding of the content, such as writing explanations or identifying multiple relationships, are worth more points. Items that require synthesis of content and other student knowledge are weighted with more points as well. Some test items encourage students to use their Core Vocabulary decks as a reference source for terminology and concepts related to the test item.

Expected Answers and Model Responses

1. a, b, c, d (2 points)

2. (6 points)

a) light and heat

b) light, sound (Heat and motion also should be considered correct.)

c) motion, sound

d) motion, sound (Heat should also be considered correct.)

e) motion, sound (Electrical energy should also be considered correct.)

f) electrical energy, light (Heat should also be regarded as correct.)

3. (4 points)

Above Average	Student response includes labels and/or clear descriptions of what is happening in the illustration. The type of energy causing the change is identified, and student clearly describes the change.
Average	Student response includes an accurate drawing with labels. The type of energy causing the change is identified, and student describes the change.
Adequate	Student response includes a drawing. The type of energy causing the change is identified, and/or student describes the change.
Inadequate	Student response includes an inaccurate or no drawing. The type of energy causing the change is not identified or is incorrect. Student does not describe the change.

4. Student response should note that the highest ramp will result in the greatest energy of motion when the ball contacts the block. When the energy is transferred to the block, it will move it the farthest. (3 points)

5. Student response should note that the lowest ramp will result in the least energy of motion when the ball contacts the block. When the energy is transferred to the block, it will move it the least distance. (3 points)

6. Student response should note that as the speed of an object increases, the amount of its energy of motion increases. (3 points)

7. Student response should include evidence that sound energy can travel over long or short distances. (3 points)

8. a, d, e, f, g (3 points)

9. The food lamps transfer light and heat energy to the food. (2 points)

10. Accept all plausible responses. Student responses providing evidence that the lamps transfer light and heat could include: the food stays hot on the counter, you can feel the heat coming from the lamp, and you can see the light with your eyes. (3 points)

11. Student response should note that the more weight in the car, the more stored energy it probably must have that is converted to motion energy. Based on this pattern, the car with more weight will knock more blocks over than a car with less weight. (3 points)

12. Student response should note that Student A is correct because the ball has stored energy while it is positioned in the teacher's hand. This stored energy transforms into motion energy when she drops the ball. (3 points)

13. Accept all plausible student descriptions of a device that automatically provides water and light to a plant. (4 points)

14. It could be tested by making sure the plant is healthy and the soil is moist. It might be improved by adjusting it to get more light or water. (4 points)

15. Maria's drip hose solution converts energy of position to energy of motion (as water runs down from the container to the planter). Maria's lamp solution converts electrical energy to light energy. (4 points)

Appendix A

Glossary

Blue words and phrases are Core Vocabulary terms for the unit, and Student Reader page numbers are listed in parentheses. **Bold-faced words and phrases** are additional vocabulary terms related to the unit that you should model for students during instruction and that are often used within the Student Reader, and these latter terms do not have specific page numbers listed. Vocabulary words are not intended for use in isolated drill or memorization.

C

cause and effect, n. is a relationship in which one thing makes another thing happen

change. v. to become different (1)

claim, n. a statement that answers a question or poses a solution to a problem

collide. v. to come together with impact (25)

collision, n. an instance of colliding (25)

contact, n. to be physically touching

conversion, n. the process of being converted or changing form (32)

convert. v. to change form (32)

D

design problem, n. a problem for which a design solution should be applied

design solution, n. the method someone develops to solve a design problem

distance, n. amount of space between two objects

E

electrical energy, n. the flow of electrons from one location to another

electricity, n. a form of energy caused by the movement of electrons

energy, n. the ability to cause change (1)

energy change, n. the conversion of one form of energy into one or more other forms of energy

energy of motion, n. the energy an object possesses while it is moving (9)

energy transfer, n. movement of energy from one object to another or from one place to another (17)

energy transformation, n. the change of one form of energy to another form of energy (32)

engineering design, n. a process used to develop a solution to a problem (36)

evidence, n. anything that helps prove or disprove an idea

F

forms of energy, n. the different forms energy can be found in, such as light or heat

H

heat, n. a form of energy caused by the movement of atoms or molecules

L

light, n. a form of energy caused by the movement of photons, which are little packets of energy

M

model, n. a helpful tool for representing an idea or explaining a process or relationship

motion, n. the process of an object changing position (7)

O

observation, n. the act of noting or viewing something

P

position, n. the place an object occupies

prediction, n. a statement about what might or will happen

R

reasoning, n. a scientific mindset that connects claims and evidence

rotate, v. to spin

S

solution, n. a method for solving a problem

sound, n. a form of energy that comes from a vibrating object (28)

speed, n. a measurement of the distance an object travels over an amount of time (13)

stored energy, n. energy that has the ability to cause change at a later time (4)

T

transfer, v. to move from one place to another

transform, v. to change the appearance or form of something; see also ***energy transformation***

V

variable, n. something that is changed or is not consistent

Appendix B

Classroom Safety for Activities and Demonstrations

In the Core Knowledge Science program (CKSci), activities and demonstrations are a vital part of the curriculum and provide students with active engagement related to the lesson content. The activities and demonstrations in this unit have been selected and designed to engage students in a safe manner. The activities and demonstrations make use of materials and equipment that are typically deemed classroom safe and readily available.

Safety should be a priority when engaged in science activities. With that in mind, observe the following safety procedures when the class is engaged in activities and demonstrations:

- Report and treat any injuries immediately.
- Check equipment prior to usage, and make sure everything is clean and ready for use.
- Clean up spills or broken equipment immediately using the appropriate tools.
- Monitor student behavior to ensure they are following proper classroom and activity procedures.
- Do not touch your eyes, ears, face, or mouth while engaging in an activity or demonstration.
- Review each step of the lesson to determine if there are any safety measures or materials necessary in advance.
- Wear personal protective equipment (e.g., safety goggles, aprons, etc.) as appropriate.
- Check for allergies to latex and other materials that students may have, and take appropriate measures.
- Secure loose clothing, hair, or jewelry.
- Establish storage and disposal procedures for chemicals as per their Safety Data Sheet (SDS), including household substances, such as vinegar and baking soda.

Copy and distribute the Student Safety Contract, found on the next page, for students to read and agree to prior to the start of the first unit so students are aware of the expectations when engaged in science activities.

Online Resources



For additional support for safety in the science classroom, follow the links in the Online Resources Guide for this unit:

www.coreknowledge.org/cksci-online-resources

Student Safety Contract

When doing science activities, I will do the following:

- Report spills, breakages, or injuries to the teacher right away.
- Listen to the teacher for special instructions and safety directions. If I have questions, I will ask the teacher.
- Avoid eating or drinking anything during the activity unless told to by my teacher.
- Review the steps of the activity before I begin. If I have questions, I will ask the teacher.
- Wear safety goggles when working with liquids or things that can fly into my eyes.
- Be careful around electric appliances, and unplug them, just by pulling on the plug, when a teacher is supervising.
- Keep my hands dry when using tools and devices that use electricity.
- Be careful to use safety equipment like gloves or tongs when handling materials that may be hot.
- Know when a hot plate is on or off and let it cool before touching it.
- Roll or push up long sleeves, keep my hair tied back, and secure any jewelry I am wearing.
- Return unused materials to the teacher.
- Clean up my area after the activity and wash my hands.
- Treat all living things and the environment with respect.

I have read and agree to the safety rules in this contract.

_____ _____/_____/_____/

Student signature and date

_____

Print name

Dear Parent or Guardian,

During science class, we want to create and maintain a safe classroom. With this in mind, we are making sure students are aware of the expectations for their behavior while engaged in science activities. We are asking you to review the safety rules with your daughter or son and sign this contract. If you have any questions, please feel free to contact me.

_____ _____/_____/_____/

Parent or guardian signature and date

Appendix C

Strategies for Acquiring Materials

The materials used in the Core Knowledge Science program (CKSci) are readily available and can be acquired through both retail and online stores. Some of the materials will be reusable and are meant to be used repeatedly. This includes equipment such as scales, beakers, and safety goggles, but also items such as plastic cups that can be safely used again. Often these materials can be cleaned and will last for more than one activity, or even one school year. Other materials are classified as consumable and are not able to be used more than once, such as glue, baking soda, and aluminum foil.

Online Resources



The Material Supply List for this unit's activities can be found online. Follow the links in the Online Resources Guide for this unit:

www.coreknowledge.org/cksci-online-resources

Ways to Engage with Your Community

The total cost of materials can add up for an entire unit, even when the materials required for activities and demonstrations have been selected to be individually affordable. And the time needed to acquire the materials adds up too. Reaching out to your community to help support STEM education is a great way to engage parents, guardians, and others with the teaching of science, as well as to reduce the cost and time of collecting the materials. With that in mind, the materials list can be distributed or used as a reference for the materials teachers will need to acquire to teach the unit.

Consider some of the following as methods for acquiring the science materials:

- School Supply Drive—If your school has a supply drive at any point in the year, consider distributing materials lists as wish lists for the science department.
- Open Houses—Have materials lists available during open houses. Consider having teams of volunteers perform an activity to show attendees how the materials will be used throughout the year.
- Parent Teacher Organizations—Reach out to the local PTO for assistance with acquiring materials.
- Science Fair Drive—Consider adding a table to your science fair as part of a science materials drive for future units.
- College or University Service Project—Ask service organizations affiliated with your local higher education institutions to sponsor your program by providing materials.
- Local Businesses—Some businesses have discounts for teachers to purchase school supplies. Others may want to advertise as sponsors for your school/programs. Usually you will be asked for verifiable proof that you are a teacher and/or examples of how their sponsorship will benefit students.

Remember: if your school is public it will be tax exempt, so make sure to have a Tax Identification Number (TIN) when purchasing materials. If your school is private, you may need proof of 501(c)(3) status to gain tax exemption. Check with your school for any required documentation.

Appendix D

Advance Preparation for Activities and Demonstrations

Being properly prepared for classroom activities and demonstrations is the first step to having a successful and enriching science program. Advance preparation is critical to effectively support student learning and understanding of the content in a lesson.

Before doing demonstrations and activities with the class

- Familiarize yourself with the activity by performing the activity yourself or with a team, and identify any issues or talking points that could be brought up.
- Gather the necessary materials for class usage. Consider if students will gather their materials at stations or if you will preassemble the materials to be distributed to the students and/or groups.
- Identify safety issues that could occur during an activity or demonstration, and plan and prepare how to address them.
- Review the Teacher's Guide before teaching, and identify opportunities for instructional support during activities and demonstrations. Consider other Support and/or Challenge opportunities that may arise as you work to keep students engaged with the content.
- Prepare a plan for postactivity collection and disposal of materials/equipment.

While engaged in the activity or demonstration

- Address any emergencies immediately.
- Check that students are observing proper science safety practices as well as wearing any necessary safety gear, such as goggles, aprons, or gloves.
- When possible, circulate around the room, and provide support for the activity. Return to the Teacher Guide as students work, to utilize any Support and Challenge opportunities that will make the learning experience most meaningful for your students.

After the activity or demonstration

- Use your plan for students to set aside or dispose of their materials as necessary.
- Have students wash their hands after any activity in which they could come in contact with any potentially harmful substances.

When engaging students in activities and demonstrations, model good science practices, such as wearing proper safety equipment, never eating during an investigation, etc. Good science practices at a young age will lead to students observing good science practices themselves and being better prepared as they move into upper-level science classes.

Appendix E

What to Do When Activities Don't Give Expected Results

Science activities and experiments do not always go according to plan. Microwave ovens, super glue, and X-rays are just some of the discoveries made when people were practicing science and something did NOT go according to plan. In your classroom, however, you should be prepared for what to do when activities don't give the expected results or when an activity doesn't work.

When going over an activity with an unexpected result, consider these points in discussion with your students:

- Was there an error in following the steps in order? You or the student may have skipped a step. To help control for this, have students review the steps to an investigation in advance and make a check mark next to each step as they complete it.
- Did students design their own investigation? Perhaps their steps are out of sequence, or they missed a step when performing the activity. Review and provide feedback on students' investigation plans to ensure the work is done in proper sequence and that it supports the lesson's Big Question.
- When measurements were taken, were they done correctly? It is possible a number was written down incorrectly, a measurement was made in error, such as wrong unit of measure or quantity, or the starting or ending point of a measurement was not accurate.
- Did the equipment or materials contribute to the situation? For example, chemicals that have lost their potency or a scale that is not measuring accurately can contribute to the success or failure of an activity.

One of the greatest gifts a student can learn when engaged in science is to develop a curiosity for *why something happened*. Students may find it challenging or frustrating to work through a problem during an activity, but guiding them through the problem and figuring out *why* something happened will help them to develop a better sense of how to do science.



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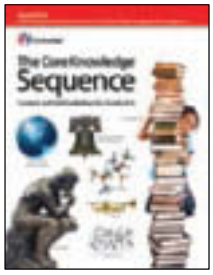
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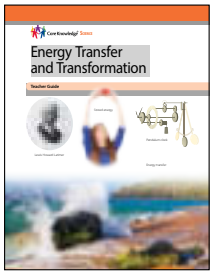
Energy Transfer and Transformation

Core Knowledge Science 4



What is the Core Knowledge Sequence?

The *Core Knowledge Sequence* is a detailed guide to specific content and skills to be taught in Grades K–8 in language arts, history, geography, mathematics, science, and the fine arts. In the domains of science, including earth and space, physical, and the life sciences, the *Core Knowledge Sequence* outlines topics that build systematically grade by grade to support student learning progressions coherently and comprehensively over time.



For which grade levels is this book intended?

In general, the content and presentation are appropriate for readers from the middle to upper elementary grades. For teachers and schools following the *Core Knowledge Sequence*, this book is intended for Grade 4 and is part of a series of **Core Knowledge SCIENCE** units of study.

For a complete listing of resources in the
Core Knowledge SCIENCE series,
visit **www.coreknowledge.org**.

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Core Knowledge SCIENCE™

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